

SWARNIM SHARMA

B.TECH INFORMATION TECHNOLOGY & MATHEMATICAL INNOVATION

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EDUCATION

Undergraduate (IT & MI) : Cluster Innovation Centre, University of Delhi | Expected 2026 | CGPA: 8/10

Intermediate (HSC) : Akash Public School | 2022 Percentage : 77.4/100

Matriculation (SSC) : Oxford Senior Secondary School | 2020 Percentage : 93/100

SKILLS

Programming Languages :	Core Skills :	Others :	Soft Skills:	Tools/Platforms:
• C/C++ • Python • MATLAB • JAVA • R-statistical computing • MY-SQL	• ML/DL • Software Automation • Neuroimaging (EEG, fMRI) • Generative-AI(LLM): framework and inference • Prompt engineering and NLP • OCR-(Multilingual) implementation	• Neuroscience (Cognitive) • Optical Coherence Tomography • Data Processing & Analysis • Software-Hardware Integration	• Research Skills • Presentation • Communication • Leadership • Teamwork • Reverse Engineering • Workflow Optimisation	• MATLAB • Cursor • Tera Term • GitHub • Windows/LINUX

PROJECTS

- **IMAGE STEGANOGRAPHY :** Developed a hybrid decoding method for image steganography using LSB and SSIS algorithms, gaining expertise in image processing, cryptography, and steganalysis metrics like PSNR and SSIM. Successfully extracted hidden messages from encoded images.
- **MULTILINGUAL OPTICAL CHARACTER RECOGNITION -:** Developed and hosted (locally) a streamlit app, that does OCR in multiple languages (English, Hindi, Tamil, etc.) and gives formatted text as output. It can also find and highlight, any word of interest in the original input file.
- **GREEN SYTHESIS OF ZINC OXIDE NANOPAETICLES :** Learned green synthesis, characterization and interaction studies of ZnO NP's and CT-DNA using physiochemical techniques. Used FTIR (Fourier Transform Infrared Spectroscopy) and UV-Spectroscopy characterization techniques

EXPERIENCE

- **Summer Intern - Neuroscience and Neurophysiology (AINN, Amity University, Noida) :** Completed a SERB-DST funded internship at AINN, focusing on "Investigating physiological alterations in cortical signaling during retinal degeneration: Implications for vision restoration." Utilized EEG and advanced signal processing techniques to analyze cortical signaling, contributing to cutting-edge research in retinal degeneration and vision restoration. This experience enhanced my skills in neuroscience and signal processing. [Jun-Jul, 2024]
- **Research Intern - Computer Science and Engineering (Parimal lab, IIT-R) :** Worked as a research intern remotely accessing their system in Parimal lab, IIT Roorkee. Designed and Implemented a ML model to analyse EEG signals to classify attention states during meditation. The process involved processing large EEG dataset through artifact removal, extracting relevant features and applying predictive modelling techniques. [Nov-Feb, 2024'25]
- **Part-time AI Applications Engineer - Generative AI in EdTech (Mythyaverse pvt. ltd.) :** Worked as an AI application engineer, supporting EdTech consultancy for "Extramarks". Using Python-based LLM frameworks (Gemini reasoning models and OpenAI o1-mini), procedurally generated age-appropriate and engaging study materials for students of CBSE and ICSE board in science and non-science subjects from class 6 to 12. Also designed and generated creative presentations for teachers using high compute reasoning models, catering to their requirements and requests for various in-classroom and out-classroom study topics. [Feb-Apr, 2025]
- **Software Development and Research Summer Intern - Automation of OCT system (FLARE lab, IISc Bangalore) :** Developed software-driven automation for the Cobra 1600 OCT spectrometer to enable faster, hands-free operation. Built Python-based control scripts to run live OCT scans, adjust exposure settings, and capture high-resolution data in real time. Integrated and debugged Camera Link hardware–software communication to ensure seamless scan control. Automated data acquisition pipelines for continuous live scanning with real-time validation and logging. Applied principles of optics, spectroscopy, and diffraction to optimize scan accuracy and imaging quality. [Jun-Aug, 2025]